

Cboe Titanium Cboe Japan Equities Smart Order Router FIX Specification

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1 Introduction

1.1 Overview

SOR clients use a subset of the FIX 4.2 protocol for order entry.

It is assumed the reader is familiar with the FIX 4.2 protocol as described at http://www.fixprotocol.org. This document describes the differences between the Cboe implementation and the FIX 4.2 standard.

1.2 Data Types

1.2.1 Timestamps

All FIX timestamps are UTC as per the FIX standard. SOR clients are expected to synchronize their clocks with an external time source.

Timestamps in messages from Cboe Japan will have microsecond or nanosecond formatting ("YYYY-MM-DD HH:MM:SS.ssssss" or "YYYY-MM-DD HH:MM:SS.ssssss000", respectively), depending on the configuration of port attribute "Nanosecond Timestamp Granularity".

Timestamps in messages to Cboe Japan may have any precision in UTC format but will be treated by Cboe Japan at no finer than microsecond precision.

1.2.2 Prices

SOR clients should program their systems to allow execution prices to be returned with up to four decimals for orders. Orders having invalid tick sizes will be rejected. Orders having a price greater than ¥99,999,999 are rejected.

1.3 Protocol Features

The SOR platform does not guarantee messages sent by SOR clients to the SOR platform, including through protocols such as TCP. SOR clients are responsible to monitor the status of the messages they send to the SOR platform.

1.3.1 Architecture

Each FIX order handler process allows a single TCP connection from a SOR client. Connection attempts from unknown source IP ranges will be blocked to prevent unauthorized access to FIX ports. The Cboe Japan Trade Desk should be contacted if a firm desires to connect from a new source IP range.

1.3.2 Maximum Open Order Limits

Cboe Japan limits the maximum number of open orders allowed on a FIX port to 300,000 per port. New orders will be rejected once this limit is breached until the number of open orders drops back below the limit.

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2 Protocol

2.1 Message Format

FIX messages are ASCII formatted. The SOR client will be provided with a *SenderCompId* and *SenderSubId* that must be sent on every message. The *TargetCompId* for all messages the SOR client sends is 'CSOR'. All messages the SOR client receives will have the Sender and Target fields swapped.

2.2 Sequence Numbers

Sequence numbers, both inbound and outbound, will be reset to 1 each night during the down time.

Messages are processed in sequence order. Behind sequence messages (other than Sequence Reset – Reset) cause immediate logout. Ahead of sequence messages (other than a Resend Request) trigger a message recovery via a Resend Request.

2.3 Version Compatibility

Cboe uses the FIX 4.2 session protocol.

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3 Sessions

The following session messages are supported in both directions:

Message	Type	Comment
Logon	A	Begin session (or resume a broken session).
Heartbeat	0	
Test Request	1	
Resend Request	2	
Reject	3	Malformed message or improper session level handling.
Sequence Reset	4	Both Gap Fill (*GapFillFlag*=Y) and Reset.
Logout	5	Used to gracefully close session.

3.1 Connectivity and Certification

IP Address	Address to connect to.	Supplied by Cboe.
TCP Port	Port to connect to.	Supplied by Cboe.
SenderCompID	Sent in every FIX message to Cboe.	Supplied by Cboe.
SenderSubID	Sent in every FIX message to Cboe.	Supplied by Cboe.
TargetCompID	Sent in every FIX message to Cboe.	'CSOR'
TargetSubID	Sent in every FIX message to Cboe.	'TEST' for test/certification system. 'PROD' for production.

For information on connectivity options to Cboe, refer to the Cboe Japan Connectivity Manual.

All new SOR clients must pass certification before they will be permitted to enter orders in the production platform. Certain new order types and options will also require certification before they may be utilized. Contact the Cboe Japan Trade Desk for more information.

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3.2 Logon

Tag	Field Name	Req'd	Description
35	*MsgType*	Y	A
108	*HeartbeatInterval*	Y	SOR client Heartbeat Interval (in seconds).

The Logon must be the first message sent by the SOR client after the TCP connection is established. EncryptMethod is ignored (FIX level encryption is not supported).

Cboe will wait one second after a Logon is received to ensure no Resend Request messages are in flight from the SOR client. A Heartbeat will be sent to indicate the one-second wait period has ended. **SOR clients should not send any orders prior to receiving this first Heartbeat from Cboe.**

The IP Address of the SOR client, the *SenderCompId*, *SenderSubId, TargetCompId*, and *TargetSubId* ('TEST'/'PROD') will be validated. If validation fails, the connection will be dropped without a Reject (to avoid corrupting the SOR client's sequence in the case that the SOR client merely mistakenly connected to the wrong port).

If connection is unexpectedly broken, upon reconnection the SOR client may receive a login reply with a sequence number greater than expected. This means in-flight messages were missed (likely important Execution Reports). The SOR client should issue a Resend Request to retrieve the missed messages.

Similarly, Cboe will issue a Resend Request to the SOR client for messages it missed. The SOR client may wish to send gap fill messages in place of new orders to avoid re-submission of potentially stale orders.

HeartbeatInterval must be specified by the SOR client in the Logon message. This value will be clamped between 5 and 300 seconds and returned in the logon reply message. We recommend using as low a value as the reliability and latency of your connectivity will allow.

3.3 Heartbeat

Tag	Field Name	Req'd	Description
35	*MsgType*	Y	0
112	*TestReqID*	N	Required in response to a Test Request.

A Heartbeat message should be sent if the agreed upon *HeartbeatInterval* has elapsed since the last message sent. If any message has been sent during the preceding *HeartbeatInterval* a Heartbeat message need not be sent.

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3.4 Test Request

Tag	Field Name	Req'd	Description
35	*MsgType*	Y	1
112	*TestReqID*	Y	Auto-generated request ID.

If a *HeartbeatInterval* + 1 seconds have elapsed since the last message received, a Test Request should be issued. If another *HeartbeatInterval* + 1 seconds elapse without receiving a message, the TCP connection should be dropped. This ensures a broken TCP connection is detected even if the TCP stack does not notice (this has been observed to happen in WAN environments, particularly when a VPN is involved).

3.5 Resend Request

Tag	Field Name	Req'd	Description
35	*MsgType*	Y	2
7	*BeginSeqNo*	Y	Auto-generated request ID.
16	*EndSeqNo*	Y	0 means + infinity

A Resend Request message should be processed even if it is received ahead of sequence. Only after resending the requested range (all marked *PossDup*='Y', including any gap fills) should Resend Request be issued in the opposite direction.

As discussed in the FIX 4.2 specification, it is possible to send an open or closed sequence range in a Resend Request (an open range uses sequence zero as the *EndSeqNo*). Cboe will honor either type of request but will always issue Resend Requests with a closed sequence range.

3.6 Reject

Tag	Field Name	Req'd	Description
35	*MsgType*	Y	3
45	*RefSeqNum*	Y	*MsgSeqNum* of rejected message.
371	*RefTagID*	N	
372	*RefMsgType*	N	
373	*SessionRejectReason*	N	
58	*Text*	Y	

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Session level rejects are used to indicate violations of the session protocol or missing (or bogus) fields. These are to be expected during development and certification while the SOR client is being adapted for Cboe but should be extremely rare in production. Application layer rejects (like Order Reject and Cancel Reject) are normal.

3.7 Sequence Reset

Tag	Field Name	Req'd	Description
35	*MsgType*	Y	4
36	*NewSeqNo*	Y	*Next expected sequence number.*
123	*GapFillFlag*	N	Sequence Reset – Gap Fill messages (*GapFillFlag*='Y') must be received in sequence. Any messages (including any Gap Fills) sent in response to a Resend Request should have PossDup='Y'. Sequence Reset – Reset (*GapFillFlag* not 'Y') is used only as a last resort, and always by human intervention, to allow an otherwise hopelessly confused session to be resumed. In these cases, all chance at automatic message recovery is lost.

3.8 Logout

Tag	Field Name	Req'd	Description
35	*MsgType*	Y	5
58	*Text*	N	*Indicates reason for Logout.*

Either side may issue a Logout to gracefully close the session. The side issuing the Logout should process messages normally until it sees the logout reply, and then break the TCP connection.

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4 FIX Messages

4.1 Standard Message Header

Tag	Field Name	Req'd	Description
8	*BeginString*	Y	FIX.4.2 Must be first field in message.
9	*BodyLength*	Y	Length of message following *BodyLength* field up to and including the delimiter preceding the *CheckSum* field. Must be second field in message.
35	*MsgType*	Y	Must be third field in message. 0 = Heartbeat 1 = TestRequest 2 = ResendRequest 3 = Reject 4 = SequenceReset 5 = Logout 8 = ExecutionReport 9 = OrderCancelReject A = Logon D = NewOrderSingle F = OrderCancelRequest G = OrderCancelReplaceRequest B = News
49	*SenderCompID*	Y	ID of sender: Assigned by Cboe for messages sent to Cboe. (*TargetCompID* for messages from Cboe)
50	*SenderSubID*	Y	Sub ID of sender: Assigned by Cboe for messages sent to Cboe. (*TargetSubID* for messages from Cboe)
56	*TargetCompID*	Y	ID of destination: 'CSOR' for messages sent to the Cboe Japan SOR. (*SenderCompID* for messages from Cboe)

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57	*TargetSubID*	Y	Sub ID of destination: 'TEST' for messages sent to Cboe test/certification system. 'PROD' for messages sent to Cboe production system. (*SenderSubID* for messages from Cboe)
34	*MsgSeqNum*	Y	Sequential sequence number for session.
43	*PossDupFlag*	N	N = (Default) Y = Indicates a message resend and should be used during FIX Replay. New orders with a value of 'Y' will be rejected by Cboe unless port attribute "ClOrdID Session Uniqueness" is set to "Session Unique, No Duplicate Detection".
52	*SendingTime*	Y	UTC date-time the message was sent.
97	*PossResend*	N	N = Original transmission (Default) Y = Possible resend See *PossResend* description in each message type of section "4.3 Order Protocol – SOR Client to Cboe" for additional detail.
122	*OrigSendingTime*	N	For messages with *PossDupFlag*='Y', indicates time the message was first sent.

4.2 Standard Message Trailer

A standard trailer terminates each message.

Tag	Field Name	Description
10	*CheckSum*	Modulo 256 *CheckSum* of all characters in message up to and including the delimiter preceding the *CheckSum* field. Three digits with leading zeroes if necessary.

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4.3 Order Protocol – SOR Client to Cboe

SOR clients use the messages in this section to enter, amend, and cancel orders.

4.3.1 New Order Single

For direct routed orders, additional tags will be forwarded to the destination.

Tag	Field Name	Req'd	Description
35	Standard Message Header	Y	*MsgType*='D'
97	*PossResend*	N	N = **(default)** indicates a new order. Y = indicates an application level resend. If port attribute "ClOrdID Session Uniqueness" is set to "Session Unique, No Duplicate Detection", the message will be processed. Otherwise, the message will be logged but effectively ignored.
1	*Account*	N	Used by a SOR client to provide a broker cross reference. Up to 10 valid characters (ASCII range 32-126).
11	*ClOrdId*	Y	Day-unique ID chosen by SOR client. 20 characters or less. Characters in ASCII range 33-126 are allowed, except for double-quote (34), comma (44), "at" symbol (64), semicolon (59), and pipe (124). If the *ClOrdId* matches a live order it will be rejected as duplicate (unless *PossResend* (97) ='Y', see above). **Note: Cboe does not fully enforce the uniqueness of *ClOrdID* values. However, we *strongly* recommend keeping *ClOrdID* values day unique.**
38	*OrderQty*	Y	Number of shares for order, 1 to 999,999,999.
40	*OrdType*	Y	1 = Market 2 = Limit 5 = Market On Close (route to TSE only) B = Limit on Close (route to TSE only) I = Funari
44	*Price*	Y	Limit price of the order. Required if *OrdType*(40) is Limit (2), Limit on Close (B), or Funari (I). Ignored if *OrdType* is Market (1).

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47	*OrderCapacity*	Y	The capacity for the order. A = Agency P = Principal
54	*Side*	Y	1 = Buy 2 = Sell 5 = Sell Short (SOR client affirms ability to borrow) 6 = Sell Short Exempt
55	*Symbol*	Y	Symbol (upper case alphanumeric) confirming to the QUICK symbology.
59	*TimeInForce*	N	0 = DAY (default) 2 = At the Opening 3 = IOC (portion not filled immediately is cancelled).
60	*TransactTime*	Y	Time order initiated/released. Required by FIX 4.2, but not used by Cboe.
544	*CashMargin*	N	Indicates the type of cash or margin order. 1 = Cash (default) 2 = Margin Open 3 = Margin Close **Rejected** if any other value is specified.
847	*TargetStrategy*	Y	1000 = Cboe Japan Routing Strategy
848	*TargetStrategyParameters*	Y	Any strategy parameter assigned specifically to the SOR client by Cboe Japan, or for direct routes any of: CHIJ = Cboe Japan Alpha XTKS = Tokyo Stock Exchange
8214	*MarginTransactionType*	N	Specifies if a margin order is a Negotiable or a Standardized. 1 = Negotiable 2 = Standardized Required if *CashMargin*(544) is 2 (Margin Open) or 3 (Margin Close), otherwise ignored. **Rejected** if any other value is specified.
	Standard Message Trailer	Y	

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4.3.2 Order Cancel Request

Request to cancel an order.

Tag	Field Name	Req'd	Description
35	Standard Message Header	Y	*MsgType*='F'
97	*PossResend*	N	N = (**default**) indicates a new cancel. Y = Indicates an application level unsolicited resend. The cancel is processed in the same manner as *PossResend*=N.
11	*ClOrdID*	Y	Day-unique ID chosen by SOR client. 20 characters or less. Characters in ASCII range 33-126 are allowed, except for double-quote (34), comma (44), "at" symbol (64), semicolon (59), and pipe (124). Duplicate order *ClOrdID*s will be rejected (or ignored if *PossResend*='Y').
37	*OrderID*	N	*OrderID* supplied by Cboe on the order acknowledgement. Optional but recommended for performance. *OrigClOrdID* must be sent if *OrderID* is not.
41	*OrigClOrdID*	N	*ClOrdID* of the order to cancel. *OrderID* must be sent if *OrigClOrdID* is not.
54	*Side*	N	**Ignored**
55	*Symbol*	N	**Ignored**
60	*TransactTime*	N	Time cancel initiated/released. Required by FIX 4.2 but not used by Cboe.
	Standard Message Trailer	Y	

4.3.3 Order Cancel/Replace Request

Request to modify an order. For direct routed orders, additional tags will be forwarded to the destination.

Requests which cannot be processed will be rejected using the `Cancel Reject` message. The `Cancel Reject` message will provide the *ClOrdID* and *OrigClOrdID* values which were specified on the `Cancel/Replace` message for identification.

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A `Cancel/Replace` should not be issued until the ack for the previous `Cancel/Replace` has been received for that order (or the acknowledgement for the first `Cancel/Replace Request`). The FIX handler will reject a new `Cancel/Replace` if it has not seen the ack for the prior `Cancel/Replace` message from the Matching Engine.

Tag	Field Name	Req'd	Description
35	Standard Message Header	Y	*MsgType*='G'
97	*PossResend*	N	N = (default) indicates a new Cancel/Replace. Y = Indicates an application level resend. The Cancel/Replace is processed in the same manner as *PossResend*=N.
11	*ClOrdId*	Y	Day-unique ID chosen by SOR client. 20 characters or less. Characters in ASCII range 33-126 are allowed, except for double-quote (34), comma (44), "at" symbol (64), semicolon (59), and pipe (124). Duplicate order *ClOrdID*s will be rejected[1] (or ignored if *PossResend*='Y').
37	*OrderId*	N	*OrderId* supplied by Cboe on the order acknowledgement. Optional but recommended for performance. *OrigClOrdID* must be sent if *OrderId* is not. In the case of multiple changes to a single order, this will be the *OrderId* from the most recent acknowledgement.
38	*OrderQty*	Y	Number of shares for order. This will modify the *OrderQty* of the current order; it does **not** directly set the remaining quantity.
40	*OrdType*	N	Not required if unchanged. Any of: 1 = Market 2 = Limit 5 = Market On Close (route to TSE only) B = Limit on Close (route to TSE only) I = Funari
41	*OrigClOrdID*	Y	*ClOrdID* of the order to replace. In the case of multiple changes to a single order, this will be the *ClOrdID* of the **most recent** accepted change.
44	*Price*	N	Limit price for limit order.
59	*TimeInForce*	N	Not required if unchanged. Any of:

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			0 = DAY (default) 2 = At the Opening 3 = IOC (portion not filled immediately is cancelled).
60	*TransactTime*	Y	Time Cancel/Replace initiated/released.
	Standard Message Trailer	Y	

[1] Cancel/Replace messages that merely reduce *OrderQty* may be overlapped if the existing *ClOrdID* is re-used.

4.4 Order Protocol – Cboe to SOR Client

4.4.1 Execution Report

This section contains all possible fields sent in Execution Reports. Not all fields are sent on each Execution Report. Additional or unexpected FIX fields on all FIX messages should be ignored.

Tag	Field Name	Req'd	Description
35	Standard Message Header	Y	*MsgType*='8'
1	*Account*	N	Copied from order.
6	*AvgPx*	Y	Average price of executions for this order weighted by trade size. Zero if *CumQty* is zero.
11	*ClOrdID*	Y	*ClOrdID* of the order being accepted, executed or rejected. -or- *ClOrdID* of the cancel or replace request. -or- *ClOrdID* of the order subject to unsolicited cancel (*OrigClOrdID* will not be present).
14	*CumQty*	Y	Cumulative quantity of shares executed for this order.
17	*ExecID*	Y	Day-unique id of execution message.
19	*ExecRefID*	N	Present only on cancel reports *ExecType* (150) = 4. 9 = Represents that the order was invalidated by TSE after Limit Allocation 1 = Represents that the order was not subject to special execution and became invalid. 0 = All other reasons
20	*ExecTransType*	Y	0 = New
31	*LastPx*	N	Price of this fill (zero for non-fills).

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			For VWAP Order Execution, this is the indicative price. For VWAP Price Restatement, this is the final VWAP trade price.
32	*LastShares*	N	Quantity of shares traded on this fill (zero for non-fills).
37	*OrderId*	Y	*OrderId* (supplied by Cboe).
38	*OrderQty*	Y	Copied from order.
39	*OrdStatus*	Y	Status of order. `0` = New `1` = Partially Filled `2` = Filled `4` = Canceled `5` = Replaced `6` = Pending Cancel `8` = Rejected `E` = Pending Replace
40	*OrdType*	N	Copied from order.
41	*OrigClOrdID*	N	ClOrdID of the order being cancelled or replaced (for a solicited `Order Cancel`, or `Cancel/Replace`, otherwise not present).
44	*Price*	N	Copied from order.
47	*OrderCapacity*	N	Copied from order.
54	*Side*	Y	Copied from order.
55	*Symbol*	Y	Copied from order.
58	*Text*	N	If present, indicates reason for reject or cancel.
59	*TimeInForce*	N	Copied from order.
60	*TransactTime*	N	UTC date-time that the transaction occurred in the SOR. (Not present if *VenueTransactTime* (20060) is present.)
76	*ExecBroker*	N	The market venue where the execution took place. Present only when *ExecType*(150)=1 or *ExecType*(150)=2. `CHIJ` = Cboe Japan Alpha `CHIS` = Cboe Japan Alpha `KAIX` = BIDS Japan `SBIJ` = Japannext J Market `XSBI` = Japannext X Market

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			XTKS = Tokyo Stock Exchange
103	*OrdRejReason*	N	Only used when *ExecType* is Rejected (8). 0 = Exchange option 1 = Unknown symbol 2 = Exchange closed 3 = Order exceeds limit 5 = Unknown order 6 = Duplicate order 8 = Stale order
150	*ExecType*	Y	Reason for this Execution Report: 0 = New (acknowledgement of new order) 1 = Partial Fill 2 = Fill 4 = Canceled 5 = Replaced 8 = Rejected
151	*LeavesQty*	Y	Quantity of shares still open for further execution. Will be zero if order is dead, otherwise will be *OrderQty* (38) – *CumQty* (14). **Note:** It is possible for *LeavesQty* to be zero when *ExecType* (150) = 5 indicating the order is dead.
198	*SecondaryOrderID*	N	OrderID as reported by the market venue. See Venue Specifics for more details.
527	*SecondaryExecID*	N	ExecID as reported by the market venue. See Venue Specifics for more details.
544	*CashMargin*	N	Copied from order if present.
8031	*PrimaryLastPx*	N	Primary exchange last traded price • if 150=1, 2, 4, or 5, the time the report is received from the venue The tag will not be present for other values of *ExecType*(150), or if there is no last traded price available from the primary exchange.
8032	*PrimaryBidPx*	N	Primary exchange best bid price at the time the order being sent to market venue.

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8033	*PrimaryAskPx*	N	Primary exchange best ask price at the time the order being sent to market venue.
8041	*PrimaryPriceFlag*	N	1 = Itayose execution when the Opening Price was fixed. 2 = Zaraba execution during Zaraba session. The order had been booked when the order was executed ("Make" orders during Zaraba session). 3 = In the case of 2 , when the order was executed as a simultaneous order. (Execution of a simultaneous order by Zaraba method) 4 = Zaraba execution during Zaraba session. The order triggered the execution ("Take" orders during Zaraba session). 5 = Itayose execution during Zaraba session (e.g., Itayose execution when a special quote or sequential trade quote was indicated.) 7 = Special Execution when the Closing Price is fixed at the end of the session. 8 = Itayose Execution when the Closing Price is fixed at the end of the session (excluding orders subject to limit allocation and special execution) 9 = In the case of 8, when the execution was the target of Limit Allocation.
8051	*RoutingDecisionTime*	N	The time when the routing decision is made to send the order to the venues. Will be returned only when an IOC order of smart routed strategy is cancelled with remaining quantity.
8052	*RoutingPhase*	N	Indicates the strategy phase of the routed order being executed in the venue. 1 = Ping 2 = Spray 3 = Conclusion
8101	*ToSTNetOrderID*	N	Order Acceptance Number in ToSTNeT Applicable for execution from BIDS Only. *ExecBroker*(76) = KAIX
8102	*ToSTNetExecutionID*	N	Execution Notice Number in ToSTNeT Applicable for execution from BIDS Only. *ExecBroker*(76) = KAIX

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8106	*BIDSTransactTime*	N	Time at which orders are matching at BIDS Applicable for execution from BIDS Only. *ExecBroker*(76) = KAIX
8214	*MarginTransactionType*	N	Copied from order if present.
20060	*VenueTransactTime*	N	Present on a cancellation due to a rejection by the remote venue. Present only if port attribute "Venue Timestamps" is enabled.
	Standard Message Trailer	Y	

4.4.2 Cancel Reject

Rejects a Cancel or Cancel/Replace request.

When a Cancel/Replace is rejected, by default the original order is left alive. A Cancel Reject should not be used as a sign that the original order has been cancelled. Even if the *CancelOrigOnReject* = *Y* option is being used a separate "unsolicited" cancel will be sent to close out the original order.

Tag	Field Name	Req'd	Description
35	Standard Message Header	Y	*MsgType*='9'
11	*ClOrdID*	Y	*ClOrdID* from the `Cancel` or `Cancel/Replace` request.
37	*OrderId*	Y	*OrderId* of order that failed to be cancelled or replaced. Represents the originating/initial order in an order chain. 'NONE' if *CxlRejReason* is not present or is Unknown (1).
39	*OrdStatus*	Y	`Cancel` or `Cancel/Replace` request: Order Status of order that failed to be cancelled or replaced. `0` = New `1` = Partially Filled `5` = Replaced `6` = Pending Cancel `8` = Rejected `A` = Pending New `E` = Pending Replace
41	*OrigClOrdID*	Y	*ClOrdID* of the order that failed to be cancelled or replaced.
58	*Text*	N	Free form text message.
60	*TransactTime*	N	The transact time of the cancel rejection by the SOR. (Not present if *VenueTransactTime* (20060) is present.)

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102	*CxlRejReason*	N	`0` = Too late to cancel. `1` = Unknown order. `2` = Broker option. `3` = Already pending cancel or pending replace.
434	*CxlRejResponseTo*	Y	This reject is in response to a: `1` = Cancel `2` = Cancel/Replace
20060	*VenueTransactTime*	N	The transact time reported by the venue rejecting the cancel request. Present only if the cancel was rejected by the venue and the port attribute “Venue Timestamps” is enabled.
	Standard Message Trailer	Y	

4.4.3 News

The News message is used to provide system-wide status information and alerts in the event of market data feed up/down and system event notification such as start and end of order acceptance.

Tag	Field Name	Req’d	Description
35	Standard Message Header	Y	*MsgType*=‘B’
148	*Headline*	Y	See description below.
33	*LinesOfText*	Y	Always 1.
56	*Text*	Y	Contains the free format text message of status information and alerts in human-readable form.
	Standard Message Trailer	Y	

4.4.3.1 Headline Format

Headline tag contents are presented as a semicolon separated sequence of “field=value” pairs. The first field is always “TYPE”. The value of TYPE may be “EVENT”, “FEED_STATUS”, or “MKT_STATUS”. The TYPE will be followed by other semicolon separated values as detailed below.

4.4.3.1.1 TYPE=EVENT

Indicates a significant event regarding order acceptance.

Additional fields present when TYPE=EVENT:

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- **EVENT:** the type of event reported. Value will be one of the following:
  - "SOOA" – Start of order acceptance. After the SOR client receives this event notification, SOR accepts new day orders from the SOR client system. IOC orders are rejected until SOMS.
  - "SOMS" – Start of morning session. After the SOR client receives this event notification, SOR accepts all new orders from the SOR client system.
  - "EOMS" – End of morning session. After the SOR client receives this event notification, SOR rejects all incoming IOC orders. After "EOMS", day orders could be queued in SOR until order could be accepted again at SOAA-TSE. IOC orders are rejected until SOAS.
  - "SOAA-TSE" – Start of order acceptance, afternoon session. After the SOR client receives this event notification, SOR accepts new day orders from the SOR client system. IOC orders are rejected until SOAS. Please note this event is in synchronous with the corresponding order acceptance event in the afternoon session from TSE.
  - "SOAS" – Start of afternoon session. After the SOR client receives this event notification, SOR accepts all new orders from the SOR client system.
  - "EOAS" – End of afternoon session. After the SOR client receives this event notification, SOR rejects all smart routed orders from the SOR client system.
- **TIME:** the time at which the event was reported, in nanosecond or microsecond resolution, as determined by the "Nanosecond Timestamp Granularity" port attribute.

Example: "`TYPE=EVENT;EVENT=SOMS;TIME=20121112-10:25:39.486853000`" (nanosecond time format)

4.4.3.1.2 TYPE=FEED_STATUS

Communicates the status of one of the market data feeds.

Additional fields present when TYPE=FEED_STATUS:

- **FEED_SOURCE**: Specifies the source of the feed for the status reported. Value will be one of the following:
  - "XTKS" – TSE Arrowhead FLEX market data feed
  - "CHIJ" – Chi-X Japan market data feed
  - "CHIS" – Chi-X Japan Select market data feed
  - "KAIX" – KAI-X market data feed
  - "SBIJ" – SBI JapanNext ITCH market data feed for J-Market
  - "XSBI" – SBI JapanNext ITCH market data feed for X-Market
- **FEED_STATE:** indicates the feed state at the time of the event. Value will be one of the following:
  - "0" – Healthy. Only this states indicates the market data feed is healthy and normal.
  - "1" – In Recovery
  - "2" – Temporary Disruption
  - "3" – Permanent Disruption
- **TIME:** the time at which the event was reported, in nanosecond or microsecond resolution, as determined by the "Nanosecond Timestamp Granularity" port attribute.

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Example: "EVENT=FEED_STATUS;FEED_SOURCE=XTKS;FEED_STATE=0;TIME=20121109-10:55:31.0147036000"

4.4.3.1.3 TYPE=MKT_STATUS

Indicates the status of one of the markets supported by the SOR.

Additional fields present when TYPE=MKT_STATUS:

- **MKT_SOURCE:** Specifies the market. Value will be one of the following:
  - "XTKS" – Tokyo Stock Exchange
  - "CHIJ" – Chi-X Japan Alpha
  - "CHIS" – Chi-X Japan Select
  - "KAIX" – KAI-X market data feed
  - "SBIJ" – SBI JapanNext J-Market
  - "XSBI" – SBI JapanNext X-Market
- **MKT_STATE:** the state of the market. Value will be one of the following:
  - "0" – Healthy. Only this state indicates access to market venue is healthy and normal.
  - "1" – Unhealthy. Venue is not accessible
- **TIME:** the time at which the event was reported, in nanosecond or microsecond resolution, as determined by the "Nanosecond Timestamp Granularity" port attribute.

Example: "EVENT=MKT_STATUS;MKT_SOURCE=XTKS;MKT_STATE=0;TIME=20121109-10:55:31.0147036"

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5 Venue Specifics

Exchange/Protocol	SecondaryOrderId(tag=198)	SecondaryExecId(tag=527)
JapanNext/FIX	OrderId(tag=37) String	TrdMatchId(tag=880) String
Cboe/BOE3	OrderId Base36(8)	ExecId Base36(8)
TSE/Arrowhead	OrderAcceptanceNumber Alphanumeric(14)	ExecutionNoticeNumber Numeric(8)

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6 Implementation Notes

6.1 Automatic Cancel on Disconnect or Malfunction

All open orders for a SOR client will be cancelled automatically if no messages have been received from the SOR client for two heartbeat intervals. This is done to prevent orders from being stuck in an unknown state in the event of telecommunications failure. SOR clients should choose their heartbeat interval carefully based on the latency and reliability of their telecommunications channel. The minimum supported interval is 5 seconds, and this is also the recommended interval if the latency and reliability of your telecommunications channel support it. Execution Reports for the automatically cancelled orders are available upon reconnection. SOR clients are responsible for rerouting orders to other market centers based on their business needs. This should be rare, but all open orders may also be cancelled in the event of a complete or partial system malfunction.

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7 FIX Port Attributes

The table below lists FIX port attributes configurable on the port or firm level. Changes to these attributes can be made by sending a written request to TradeDeskJP@cboe.com.

Attribute	Default	Description
Allow Test Symbols Only	No	Allow or disallow orders in non-test symbols.
Cancel on Disconnect	*Option #1*	Options for cancelling orders as a result of a session disconnect: • Cancel Continuous Book Orders Only (default) • Do Not Cancel Any Open Orders
Cancel on Reject	Yes	Cancels an order upon a cancel or modify reject for that order.
ClOrdID Session Uniqueness	*Option #3*	Rejects orders if the ClOrdID has previously been received on the same FIX session on the same day. 1. "Limited": reject if ClOrdID matches the ClOrdID of an open order on the same session. Slightly more performant than #2 and #3. 2. "Session Unique": reject if ClOrdID matches any ClOrdID received on the same session, including those received on cancel requests and any rejected message. 3. "Session Unique, No Duplicate Detection": same as #2, but additionally do not reject messages for having PossDup=Y or PossResend=Y, allowing them to check against ClOrdID uniqueness and all other validations.
Maximum Notional Value Per Order	No	Maximum notional value allowed on any individual order (acceptable value 1 - 10,000,000,000)
Maximum Order Size	999,999,999	Maximum number of shares allowed per order.
Nanosecond Timestamp Granularity	No	Display microsecond level timestamp granularity for *TransactTime* (60), *OrigTime* (42), and *SendingTime* (52). These tags default to microsecond granularity. If enabled, time format is YYYYMMDD-HH:MM:SS.ssssss000

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Port Order Rate Threshold	Default = 4000 msgs/sec Maximum allowed = 4000 msgs/sec	The maximum allowed message rate on the session. When the first non-administrative message is received, a one second window begins. During the second no more than 3,999 additional non-administrative messages will be allowed within that window. If the rate is exceeded all new orders in the time window are rejected, modifies are treated as cancels, and cancels are processed.
Test Symbol Order Rate Threshold	Default = 10 msgs/sec	The maximum allowed test symbol message rate on the session. Behaves identically to Port Order Rate Threshold above.
Venue Timestamps	No	Include the venue transact time as reported by the venue in Cancel Reject messages and in Execution Reports of cancellations.

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8 Contact Information

Please direct questions or comments regarding this specification to TradeDeskJP@cboe.com.

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Revision History

Document Version	Date	Description
1.0.0	22/12/15	Initial version.
1.0.1	23/02/24	Added *TransactTime* to Cancel Reject. Added *SecondaryOrderID* and *SecondaryExecID* to Execution Report. Removed unneeded port attributes "Cancel on Halt" and "Concatenate CompId and SubId". Added "Venue Cancel Reject Time" port attribute.
1.0.2	23/02/28	Added *PossResend* in Standard Message Header. Made *PossResend* the first field referenced in New Order Single. Added *ExecType* (150) = '6' and 'E' in Execution Report. Corrected *RoutingPhase* from 8502 to 8052 in Execution Report.
1.0.3	23/03/21	Added optional field *OrderId* (37) to `Order Cancel/Replace Request`.
1.0.4	23/05/01	Added "ClOrdID Session Uniqueness" port attribute.
1.0.5	23/05/22	Added detail to 1.2.1 regarding timestamps. Removed invalid references to trade reports from 1.2.2. Updated *PossDupFlag* description in Standard Message Header. Updated *PossResend* description in New Order Single and Cancel Request. Updated *ClOrdID* description in New Order Single. Removed '3' value from *OrdStatus* and *ExecType* in Execution Report. Added *OrdStatus* = '6' and 'E' in Execution Report. Added *VenueTransactTime* (20060) to Execution Report and Cancel Reject. Updated *TransactTime* description in Execution Report and Cancel Reject. Added options for port attribute "ClOrdID Session Uniqueness". Updated *PrimaryLastPx* in execution report to note it is not present when *ExecType*=New.
1.0.6	23/06/29	Removed unused values '6' and 'E' from *ExecType* (150). Added required = 'Y' to *Text* (58) in section 3.6. Changed required to 'N' and updated description for *TransactTime* (60) with respect to *VenueTransactTime* (20060). Corrected the expected *TargetCompID* value throughout to be 'CSOR'. Added *ExecRefID* (19) to Execution Report for cancellations. Changed required to 'N' for *RoutingPhase* (8052) in Execution Report.
1.0.7	23/07/25	Updated Section 3.8 "Logout" to remove reference to Cboe requesting a logout.

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		Updated Cancel/Replace to allow *OrdType* (40) to be changed. Updated Execution Report *Text* (58) description to remove invalid reference.
1.0.8	23/08/17	Removed invalid reference to “trade cancel report” in Cancel Reject. Updated *OrderId*(37) description in Cancel Reject. Updated *OrdType*(40) description of Market On Close and Limit On Close.
1.0.9	23/10/11	Removed *MinQty* from New Order Single and Execution Report. Removed the EOOA News message. Updated New Order Single *Price* field to note that it is ignored if *OrdType*(40) = Market.
1.0.10	23/12/06	Added detail for *OrdStatus*(39) in Cancel Reject.
1.0.11	24/03/04	Updated section 1.4 to include latency expectations as well as SOR clients’ responsibility to monitor the status of the messages they send to the SOR platform.
1.0.12	24/03/26	Removed “Hours of Operation” section from Introduction Update spec to include additional/updated value in ExecRefID and PrimaryPriceFlag to identify TSE Special Executions.
1.0.13	24/07/05	Updated Maximum Notional Value Per Order FIX Port Attribute
1.0.14	24/12/18	New section “Venue Specifics” added to clarify values returned in SecondaryOrderId(tag=198) and SecondaryExecId(tag=527).
1.0.15	25/01/15	Updated with Cboe Titanium branding.
1.0.16	25/06/09	Updated "Maximum Notional Value Per Order" port attribute to 1 - 10,000,000,000.

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